Decoding selective auditory attention from EEG to speech using contrastive learning with transformers

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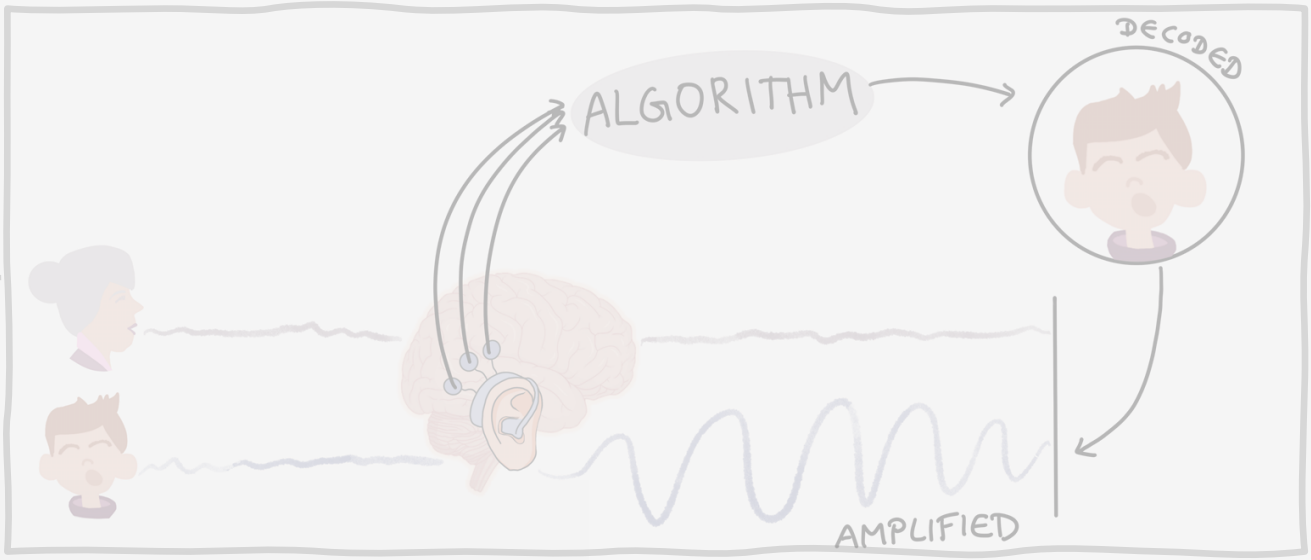
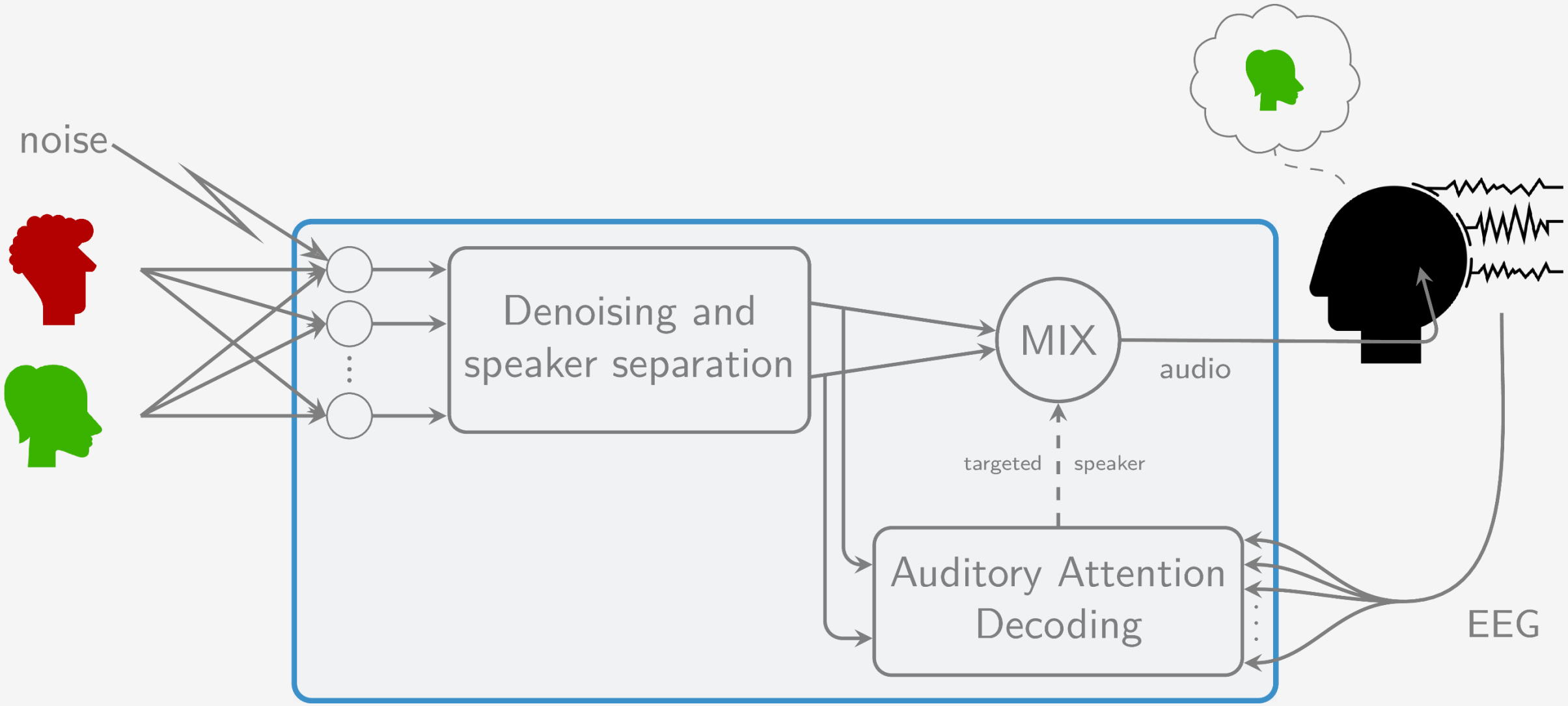
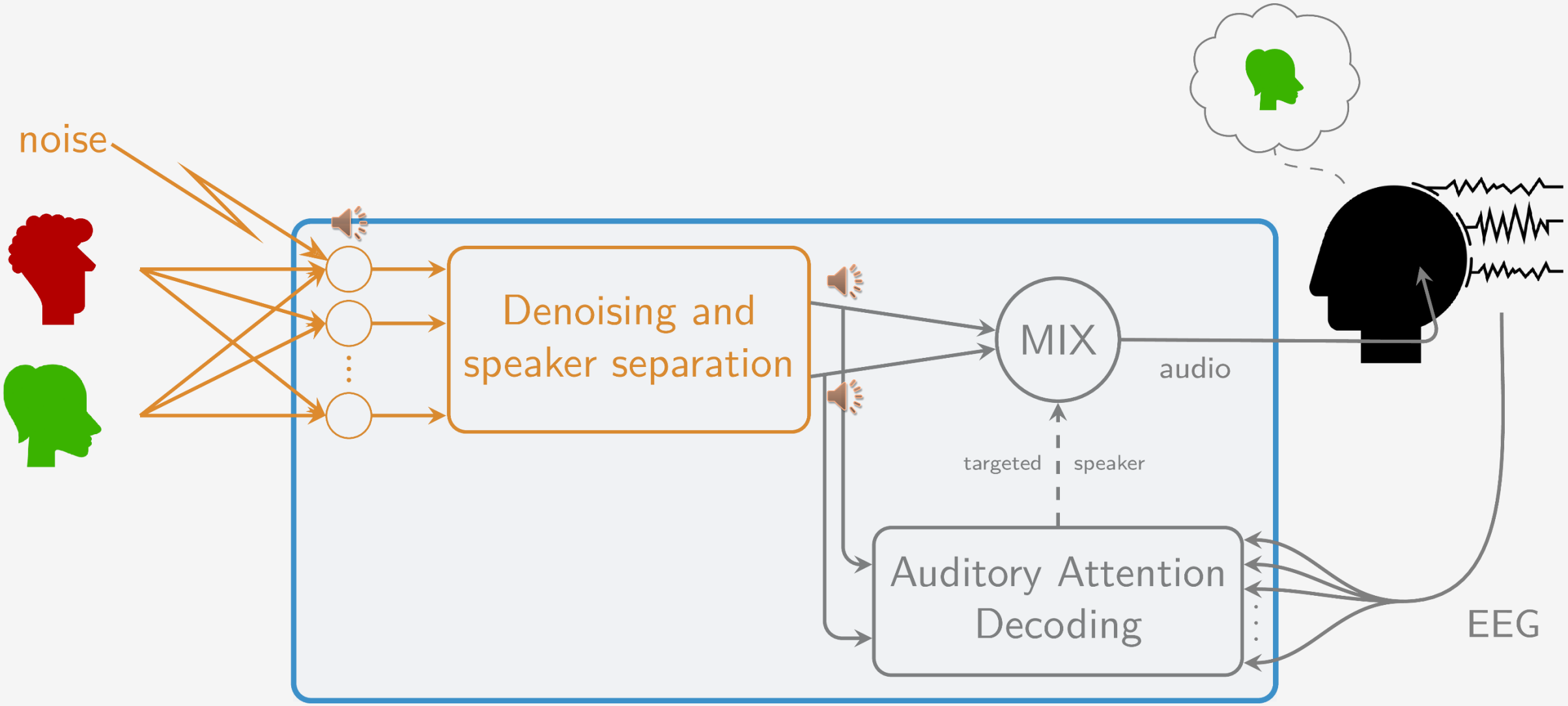
Courtesy to Debora Fieberg

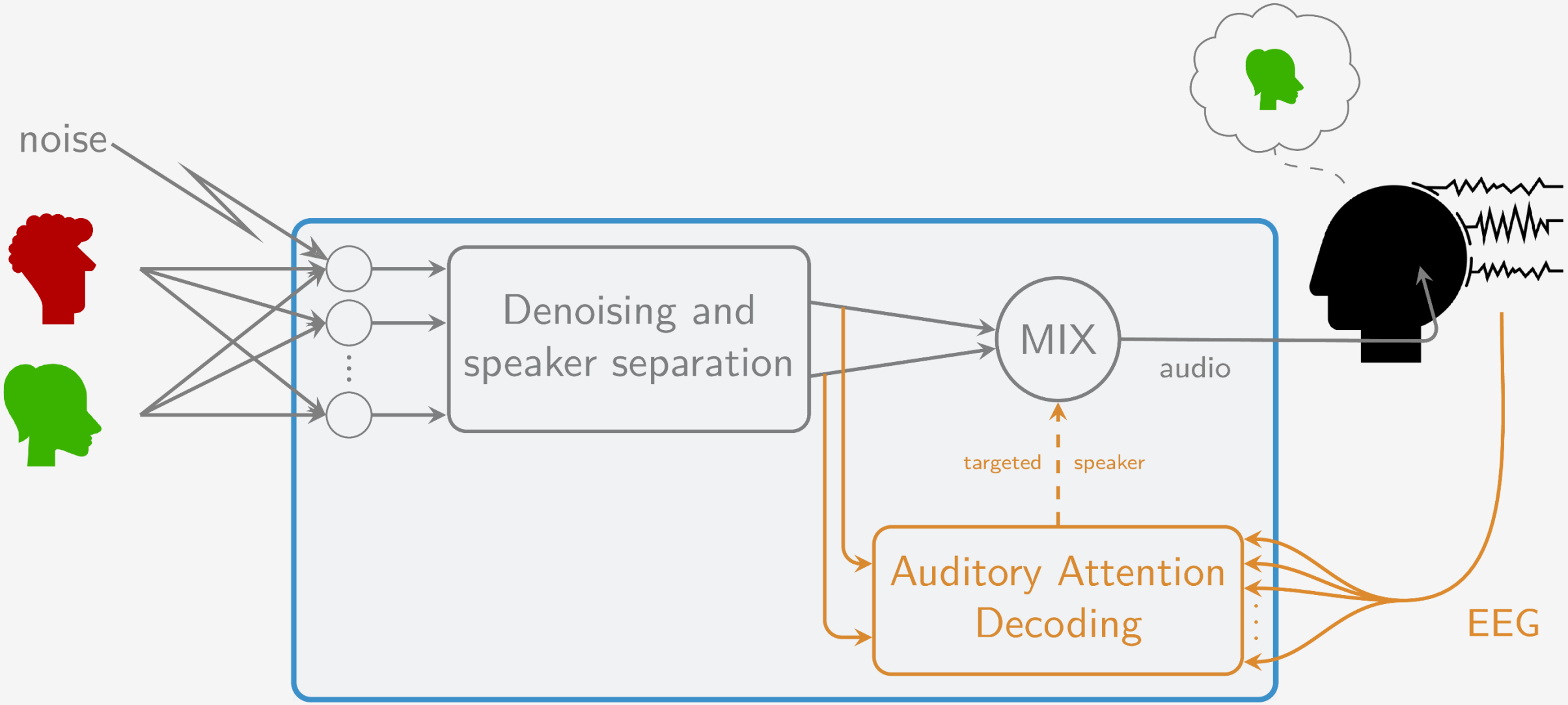


Auditory Attention Decoding (AAD) Problem



Smart neuro-steered hearing device


noise
Denoising and speaker separation
MIX
audio
targeted
speaker
Auditory Attention Decoding
EEG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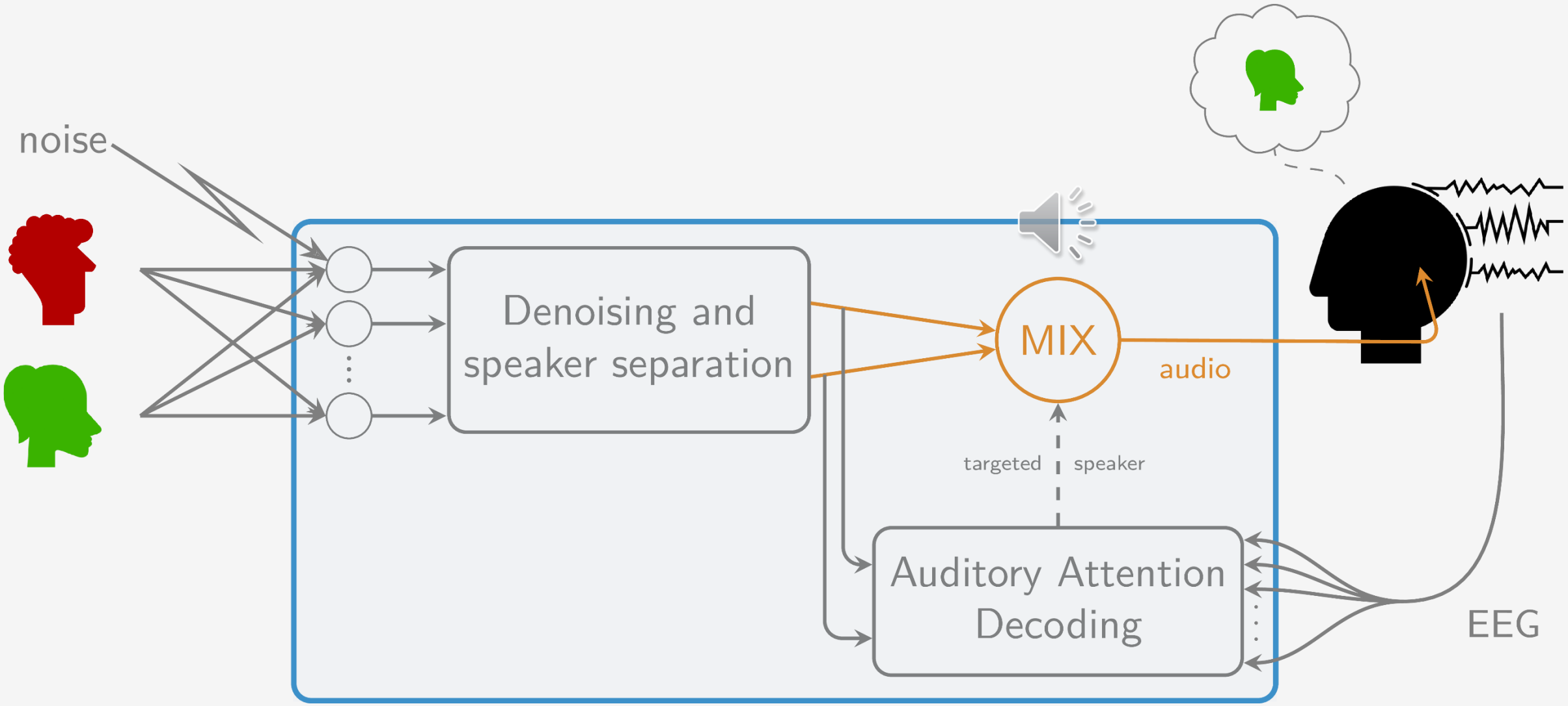
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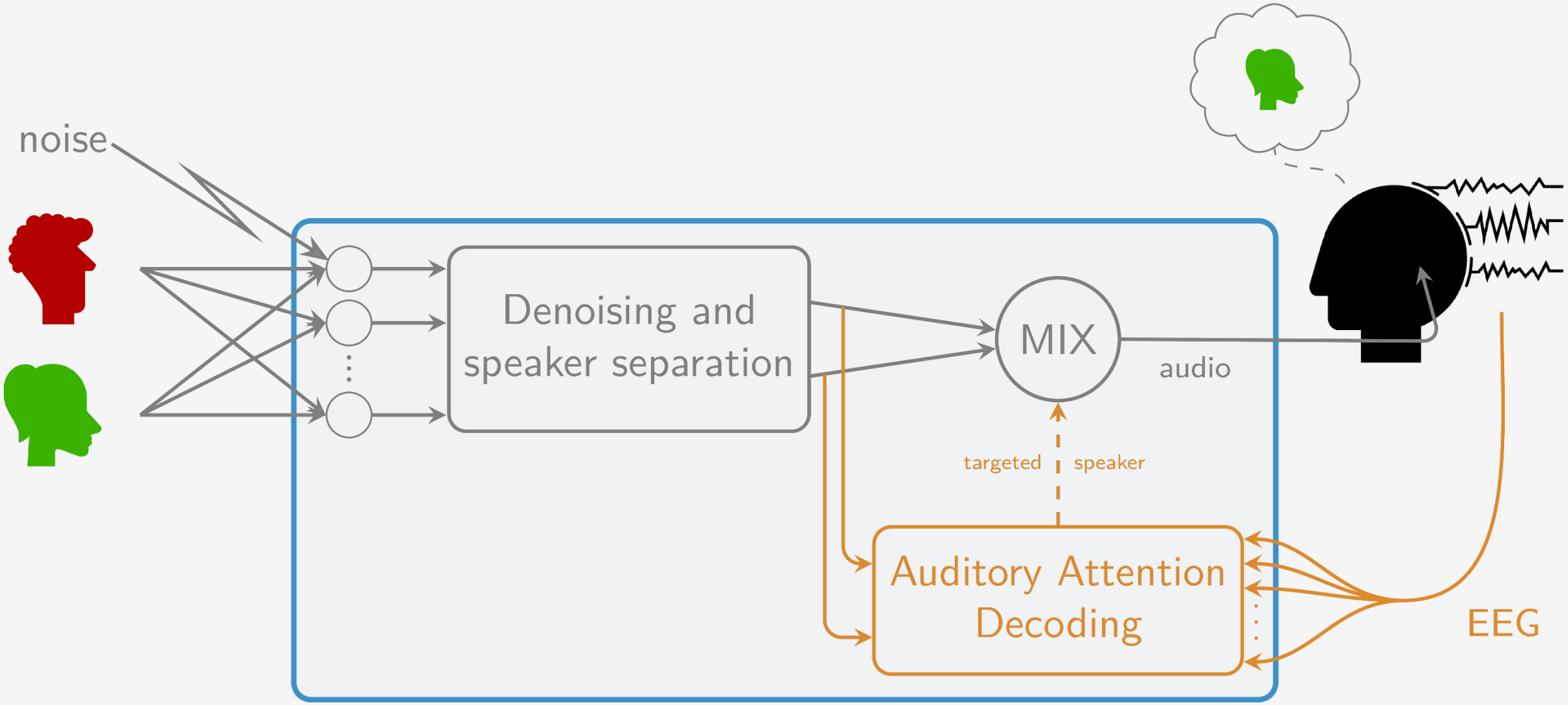
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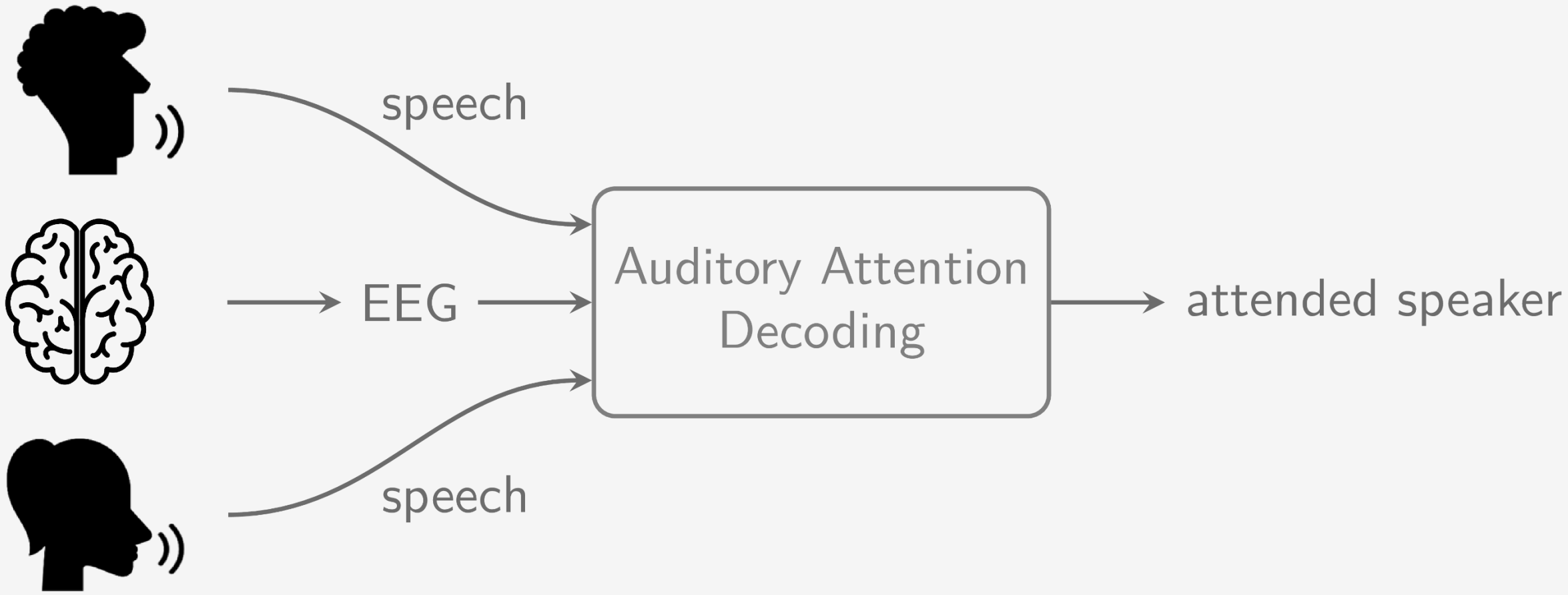
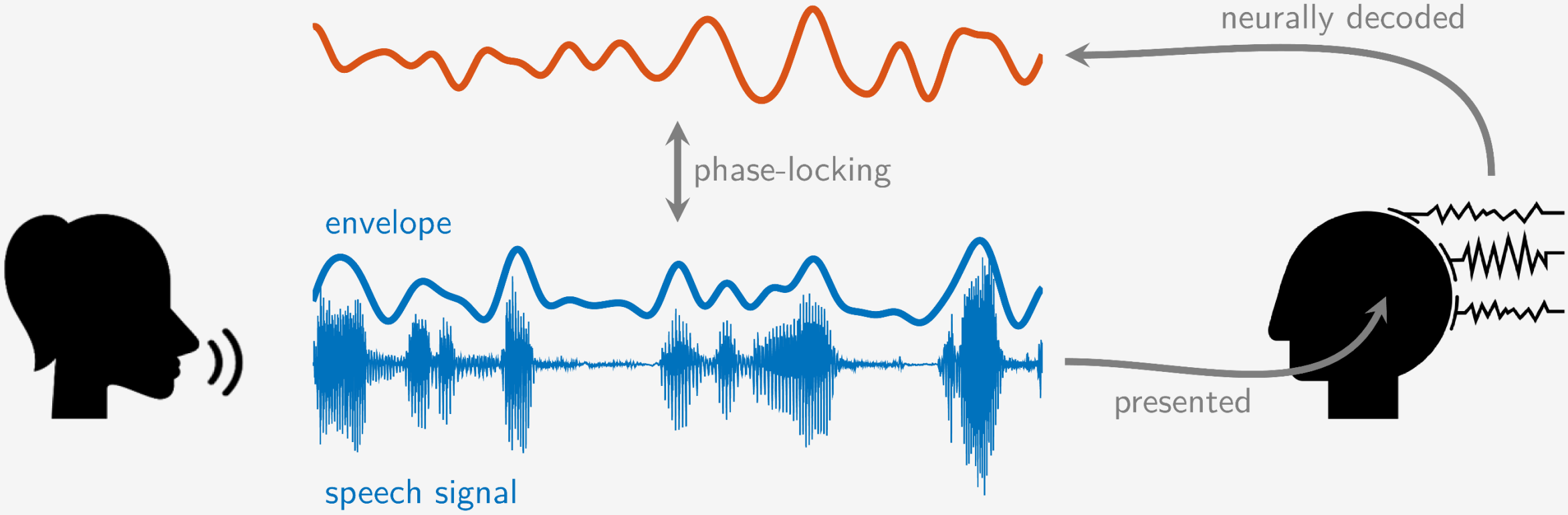
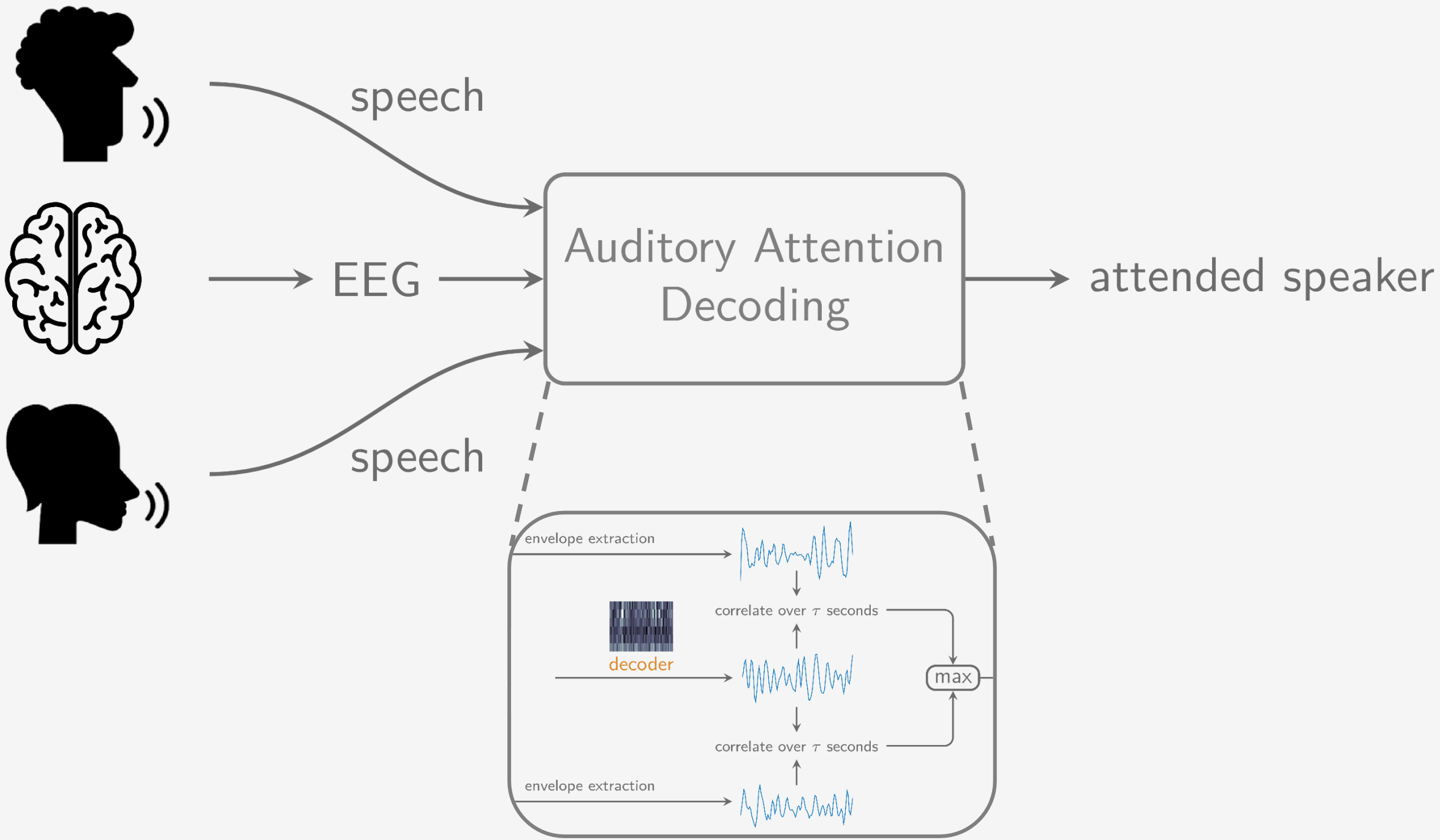
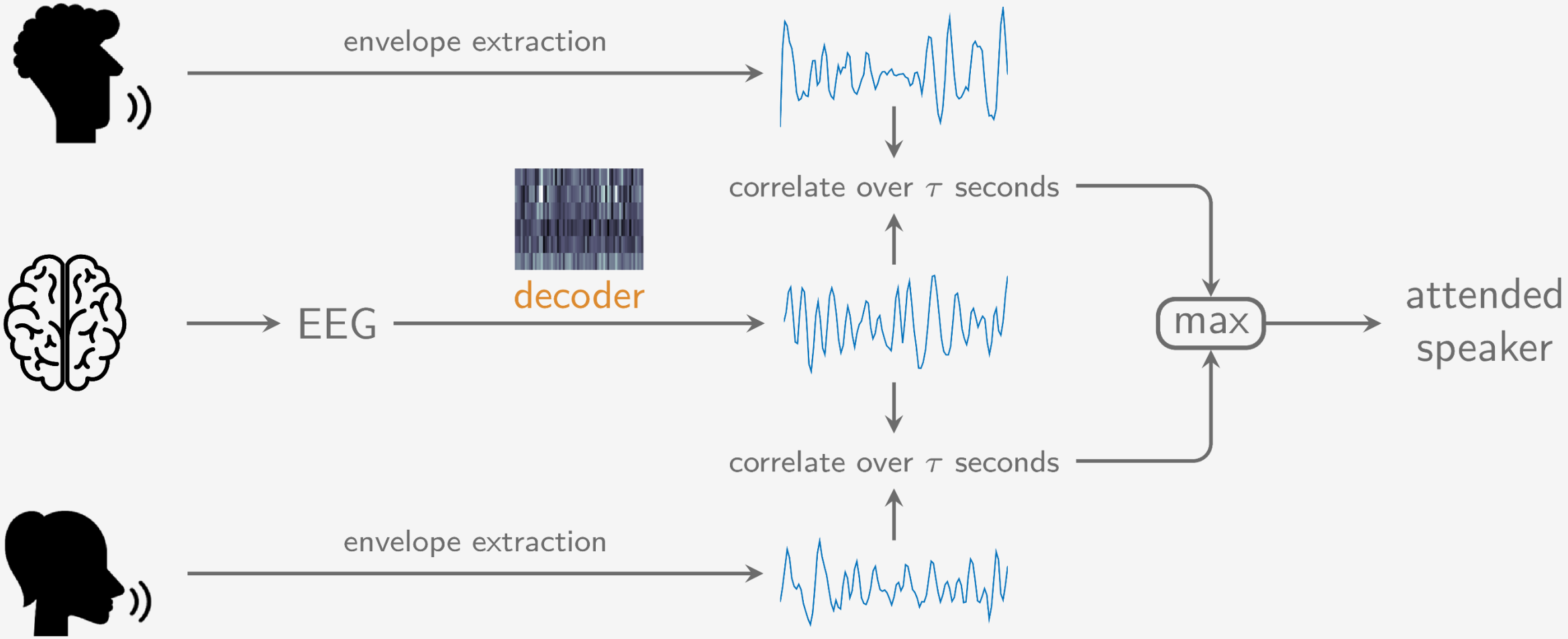

speech
EEG
speech
Auditory Attention Decoding
attended speaker

Neural tracking of speech



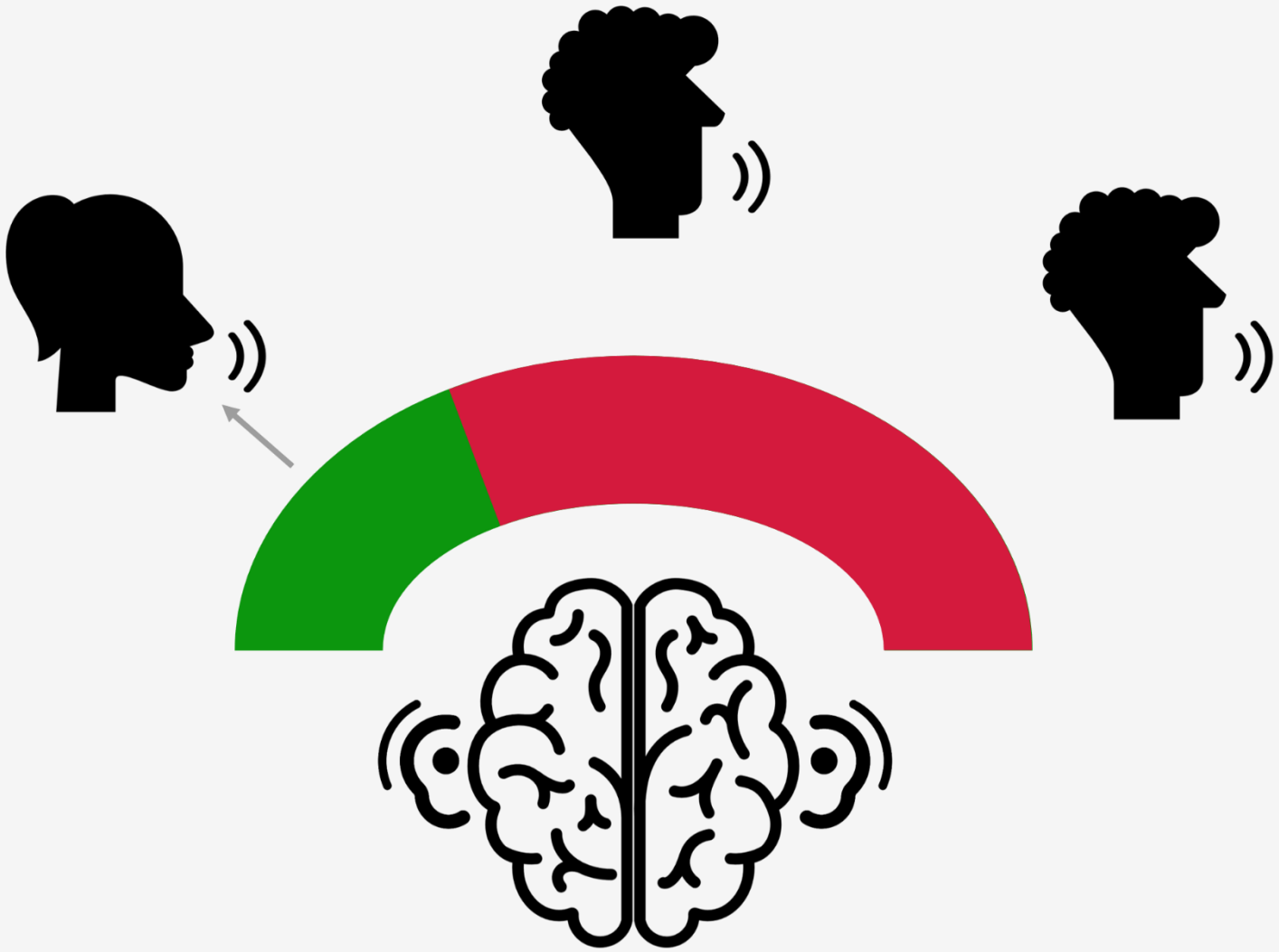

speech
EEG
speech
Auditory Attention Decoding
attended speaker
envelope extraction
correlate over τ seconds
decoder
max
correlate over τ seconds
envelope extraction

Stimulus reconstruction

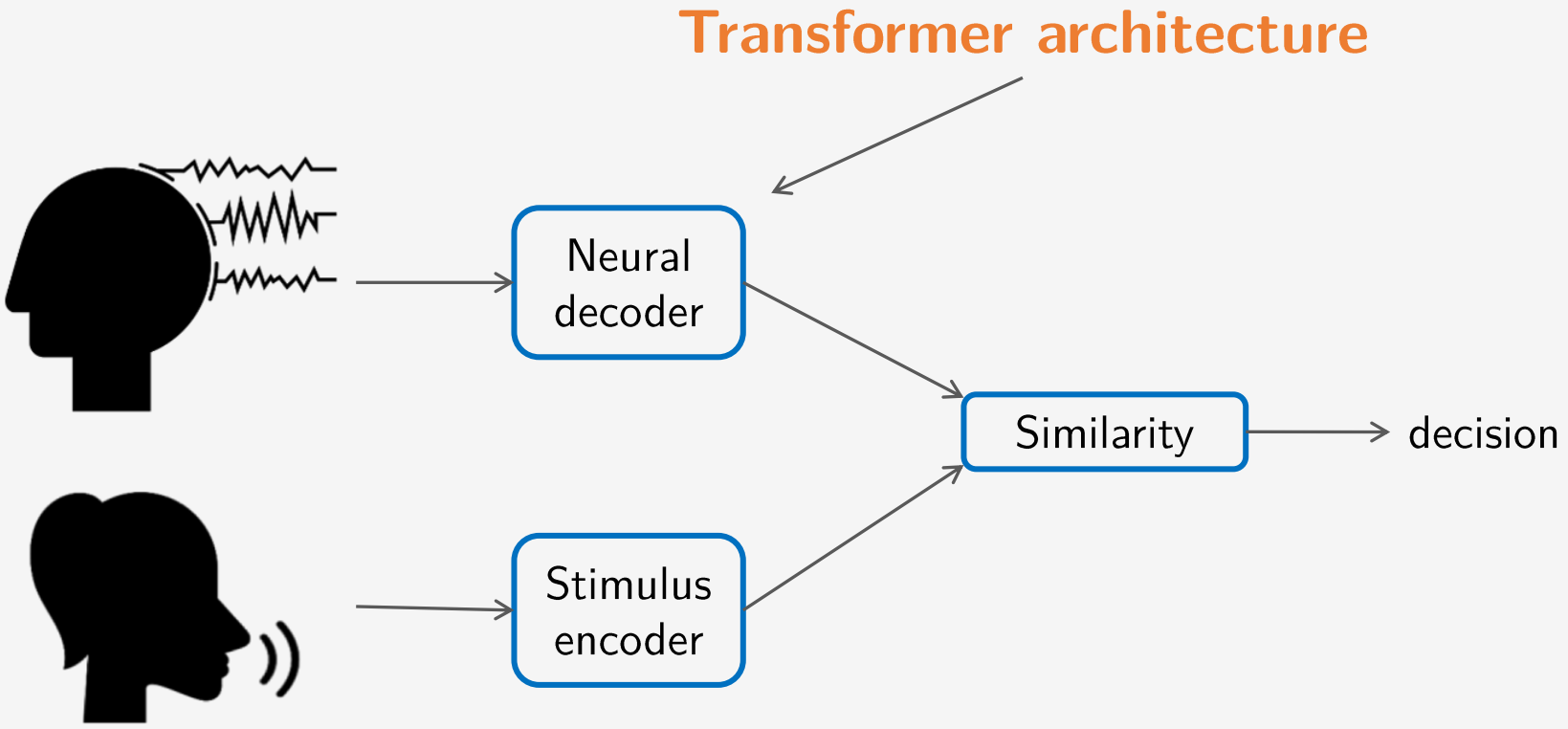


J. O'Sullivan et al, "Attentional Selection in a Cocktail Party Environment Can Be Decoded from Single-Trial EEG", *Cerebral Cortex*, vol. 25, no. 7, pp. 1697–1706, 2014

Decoding the spatial focus of auditory attention



This thesis



Objectives:

1. Design and train transformer architecture(s) for AAD using contrastive learning
2. Validate in a robust way, avoiding any type of overfitting
3. Generalize across participants and datasets

Step-by-step

1. Literature review
2. Get a feel for the data and implement a simple linear baseline
3. Design new models and train on various datasets
4. Analyze the results
5. Write thesis

You are in the driver's seat

Motivation?

What to expect from me?

Help asking the right questions

Support and not taking the lead

(bi)weekly meeting

Feedback thesis and presentations

Bridge to promotor (direct contact is limited)

You're in the driver's seat

Arrangements

Meetings: when?

Preparation of meeting: mail with agenda points

Feedback (works in **both** directions)

How to reach us: weekly meetings and via email

Code language: you can choose (suggestion: PyTorch Lightening)

Hints

Start writing in time

Ask questions

Read (but not too much)

Be creative

Limit the scope

Keep track of your time investment (time sheet)

Make a time plan (Gantt-chart, ...)